University of Cape Town



Machine Learning Assignment 1

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COURSE CODE: **STA** 5068Z

COURSE NAME: MACHINE LEARNING

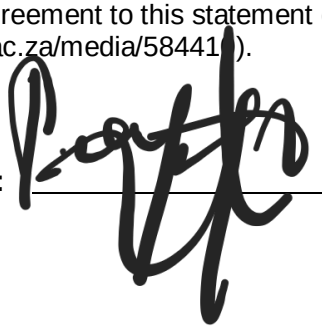
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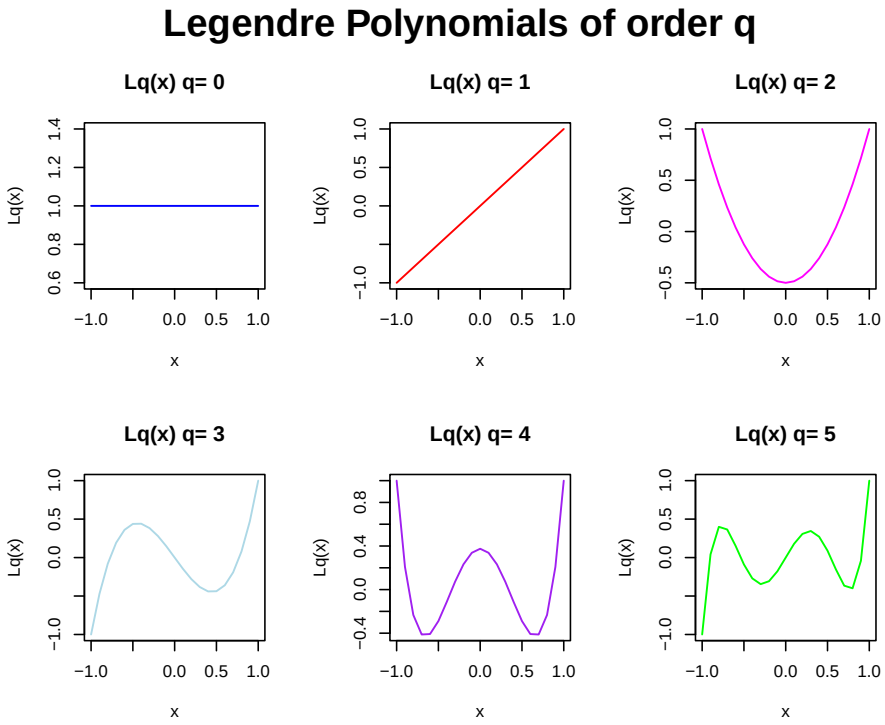
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Question 1

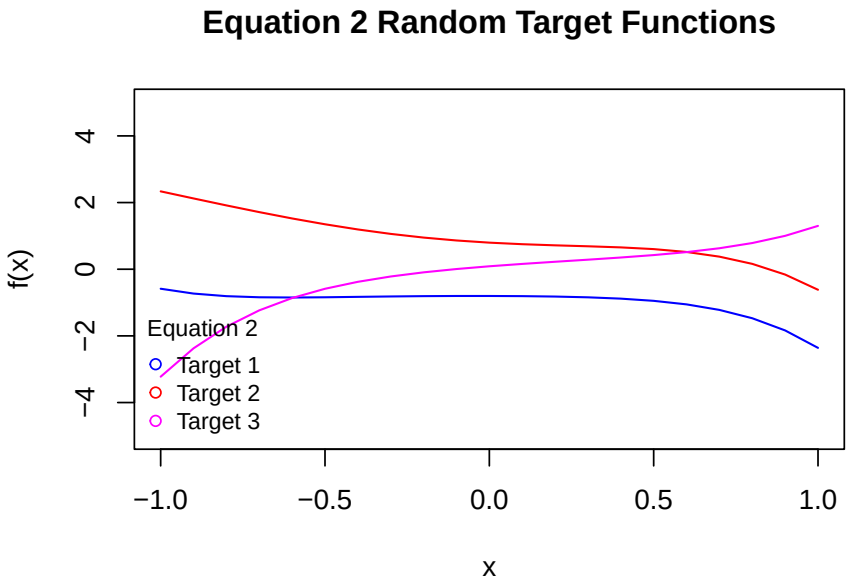
a) Legendre Polynomial Plots

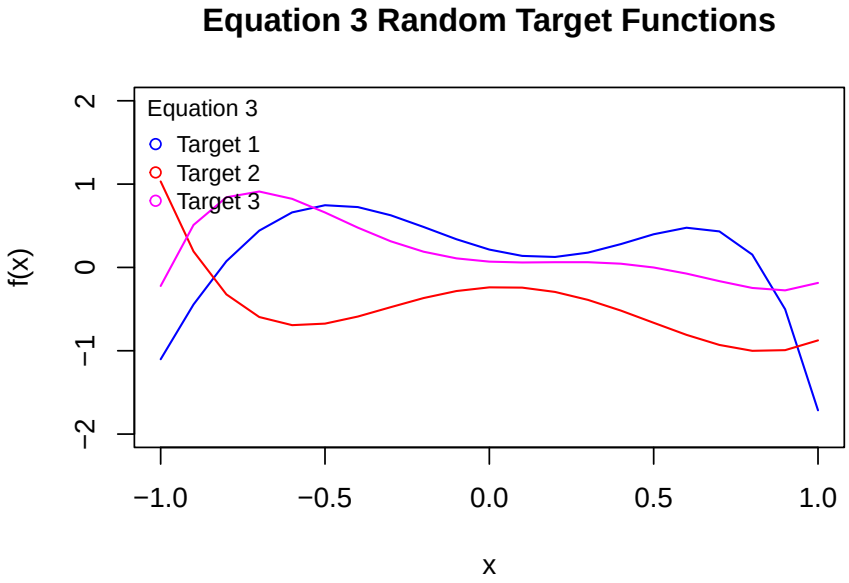


As the order of the polynomial, q, increases, the more nonlinear/complex the $L_q(x)$ becomes:

1. At q=0, $L_q(x)$ devolves to a straight horizontal line $L_q(x)$ =1 over the domain [-1,1].
2. At q=1, $L_q(x)$ starts to have some slope attached, a straight line of the form $y = 2x$.
3. At q=2 the non-linear curvature/complexity of $L_q(x)$ starts to become more visible, particularly towards the edges of the domain [-1,1]; the function approximates a parabola/quadratic function.
4. At q=3,4,5, the non-linear, polynomial curvature/complexity of $L_q(x)$ becomes more prominent with the function taking more sinusoidal shape.

b) Random Target Functions

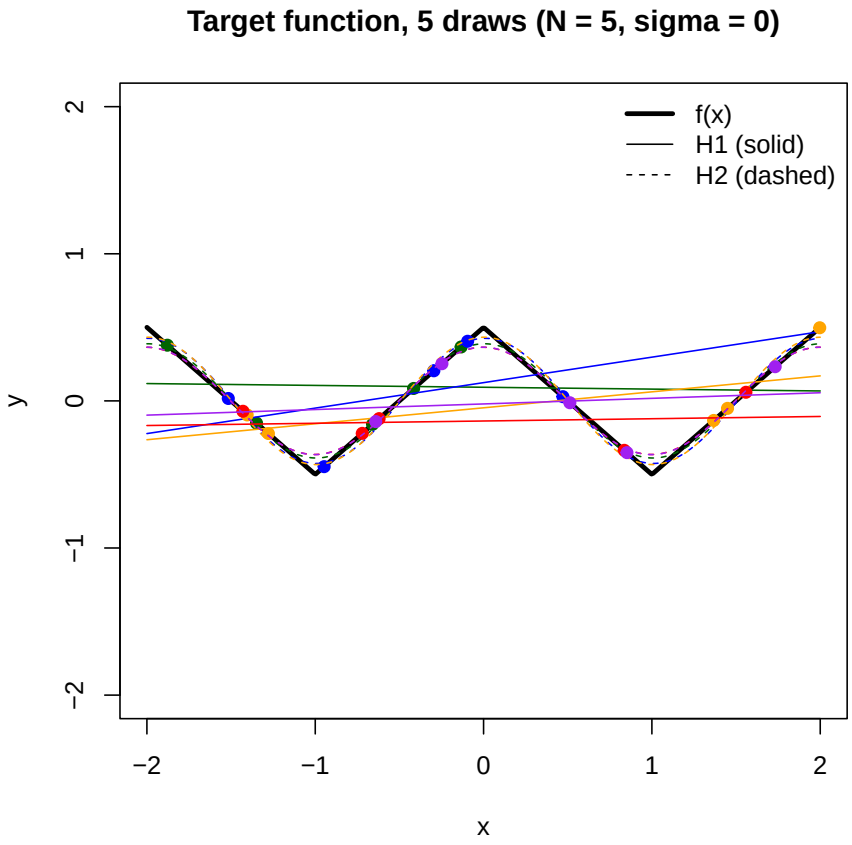




Observations:

1. $f(x) = \sum_{q=0}^{Q_f} \alpha_q x^q$: the function is smoother over the domain [-1,1], with very little curvature in between.
2. $f(x) = \sum_{q=0}^{Q_f} \beta_q L_q(x)$: there are curvatures over the same domain, with the function taking a wave-like shape.

Question 2



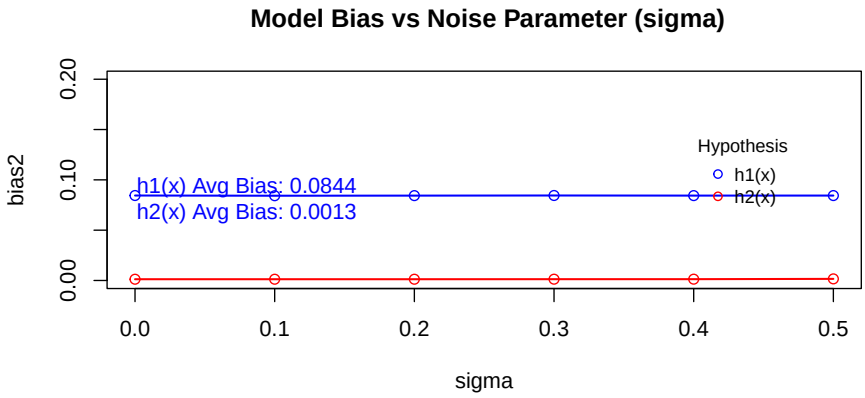
b) A Priori expectation

Given that the underlying data generating process is non-linear, the non-linear hypothesis would be more suited to fit the data's non-linear behavior and overall shape. The function has a non-linear zig-zag shape, with repeated peaks and troughs at between the interval which H2 seems to follow similarly across all 3 draws. H1 has the same number of parameters but can only produce a straight line, unable to capture the nonlinearity of the data generating process.

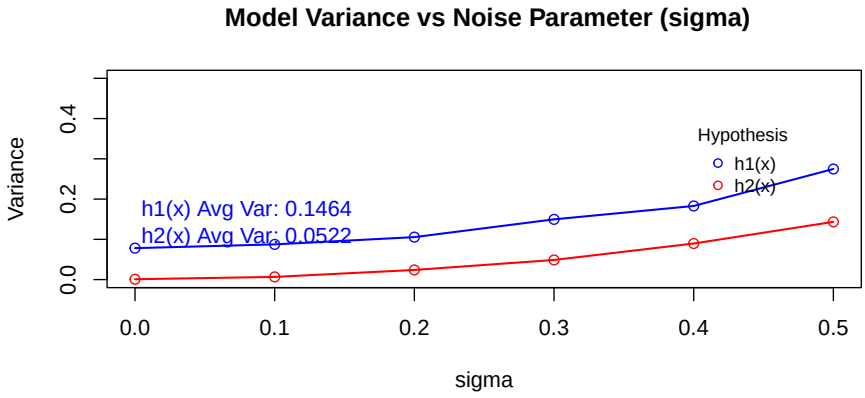
c) Bias Variance for H1 and H2

	Bias	Vari-ance	Both:Bias+Var+Noise	OOS Error	Estimate Difference
h1(x)	0.0844	0.0781	0.1626	0.1614	0.0012
h2(x)	0.0012	0.0009	0.0022	0.0020	0.0001

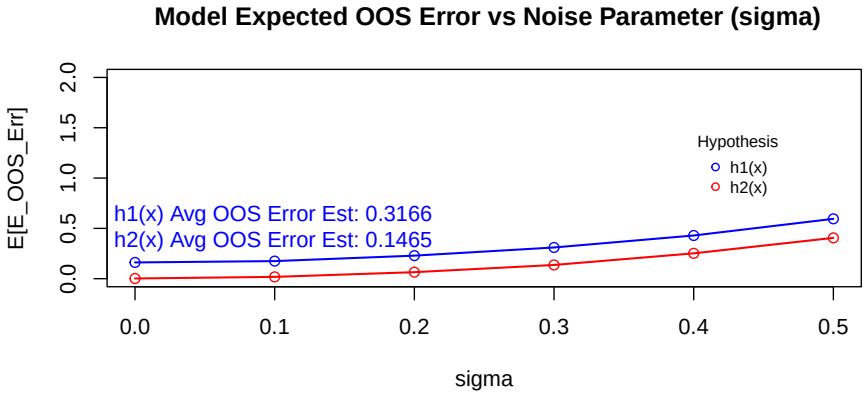
d) Bias Variance Dynamics



(a) Bias vs Noise



(b) Variance vs Noise



(c) E[OOS Error] vs Noise

Figure 1: OOS Error Estimates

e) Interpretation

Bias-Variance Dynamics

The expected out-of-sample error as it relates to the bias and variance of a model is given by

$$E[Err_{oos}] = Bias^2 + Variance + IrreducibleNoise(\varepsilon)$$

- LHS:
  - The model $h2(x)$ has better out-of-sample performance overall than the linear model $h1(x)$, averaged across 1000 simulations and varying noise parameters (average OOS error of 0.1465 vs 0.3166). It also has the lower error at every value of the noise parameter.
  - The OOS error increases for both hypotheses as the noise parameter increases, with the increase growing faster at higher noise levels: roughly 0.16 to 0.6 for $h1(x)$ and 0 to 0.41 for $h2(x)$ over the range of sigma. This is expected, since noise adds irreducible error to the OOS observations and also makes the fits less stable.
- RHS:
  - The bias² is flat across the noise parameter for both functions (0.0844 for $h1(x)$ and 0.0014 for $h2(x)$), since noise averages out across simulations and does not affect how well the average fit matches the true function.
  - The variance increases with the noise parameter for both functions, as noisier training data produces more variable fits. $h1(x)$ has the higher average variance (0.1464 vs 0.0522), and it is already non-zero at $\sigma = 0$, which comes from fitting a line to only N = 5 randomly drawn points.
  - Overall, $h2(x)$ has both lower bias and lower variance across the noise parameters. Both models have two parameters, so the difference is not complexity; the form of $h2(x)$ simply matches the target function much better than a straight line.

Model Choice

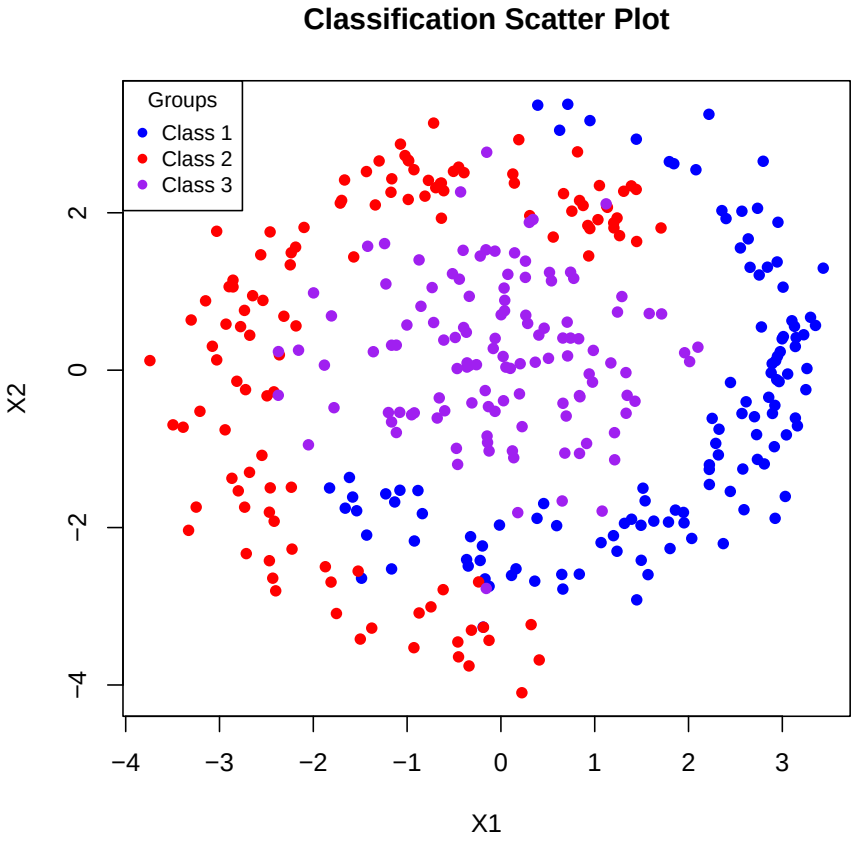
- Given the estimated OOS Error, it's better to go with the linear model as it is more parsimonious and more likely to be able to generalize to new data as observed across the above simulations.
- Rational: the combined estimate comparison $bias^2 + Var$ to $E[Err_{oos}]$ across sigmas

– $h1(x)$: its average RHS estimate of about 0.3225 is close to its LHS OOS estimate of 0.3166.

– $h2(x)$: its average RHS estimate of about 0.1453 is even closer to its LHS OOS estimate of 0.1465, so the decomposition describes the OOS behavior of $h2(x)$ very well.

Question 4

a) Data Set Plot



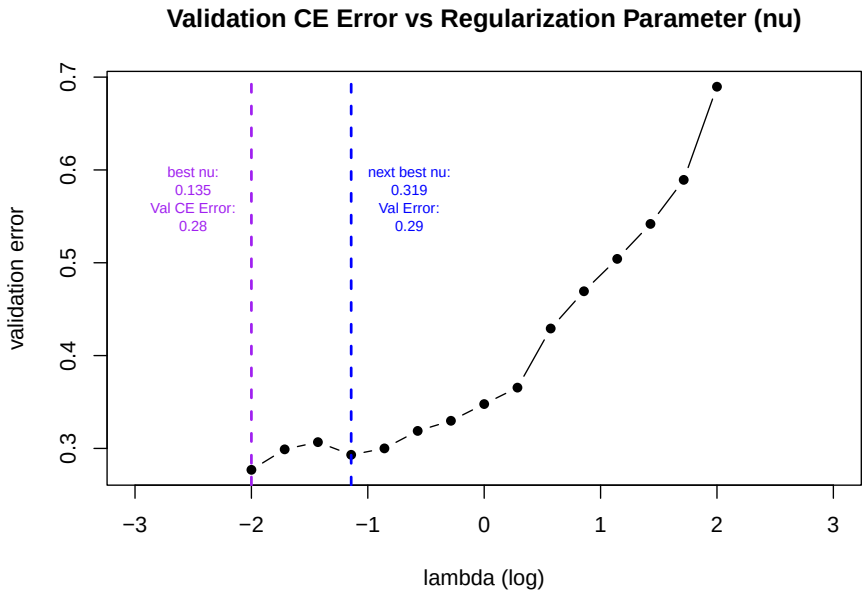
Observation: The data does not appear to be linearly separable (in the given 2D feature space) as the data does not seem to have linear boundaries between categories/classes. Instead, the different classes of responses look to be spiraled around each other. A non-linear neural network seems well suited to capture the non-linear pattern/structure embedded in the data for the given feature space.

c) Feed Forward Neural Network

Numerical Pitfalls: The computation of $log(\hat{y})$ in the objective function may cause the network to explode/ diverge given the undefined behavior of the log function at 0 (or values very close to 0), which can hamper the numerical stability of the network's computations.

d) Validation Analysis

Plot the validation error vs log (λ)

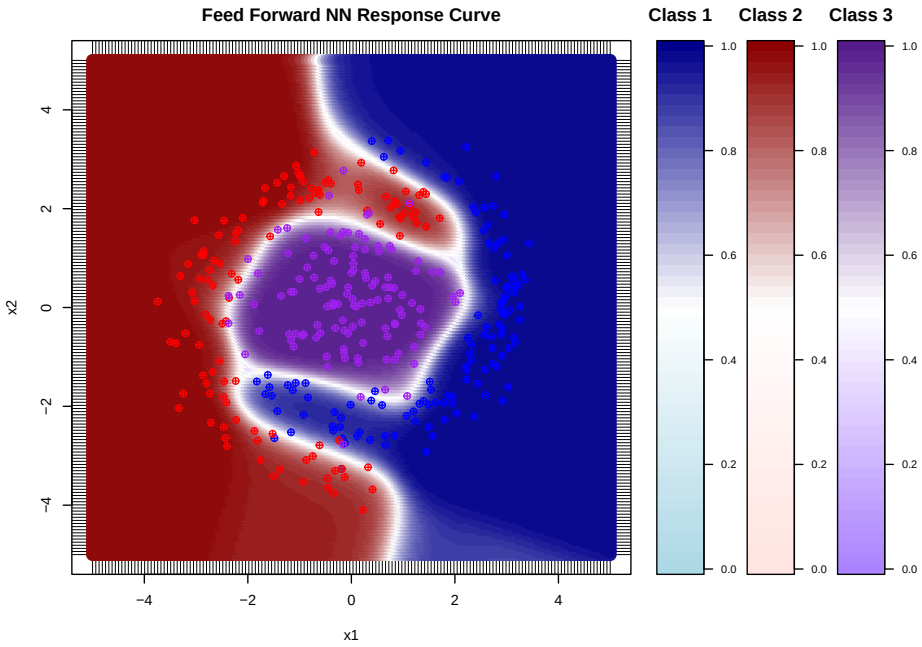


Model Parameters: number of hidden layer nodes, **m =10**

The lowest validation error is generated at $\lambda_{min} = 0.135$ with an associated validation error of 11.04, making for a less strict and more complex model (given the big budget that the sum of the model's parameters cannot exceed). The next best error to the right of this minimum is obtained at $\lambda_{1SE} = 0.319 (Val_{err} = 0.29)$, offering stricter control over the complexity of the network and thus more parsimonious model without too much of a deterioration on the error, decreasing the likelihood of the model over fitting to just the training data. **This is the better choice of regularization parameter.**

d) Response Curve

Model Hyperparameters: **m=10, nu=0.319**



The model produces three distinct, somewhat smooth regions over the feature space, each corresponding to the probability assignments for one class:

- Class 1: Most of the Class 1 observations are correctly classified. The Class 1 region covers the right side of the feature space (roughly X1 $>$ 0 to 1, depending on X2) and curves around the Class 3 region, capturing both the upper-right and lower-right arcs of blue points. The model is very confident within this region, assigning probabilities close to 1. The few misclassified observations lie near the wavy boundary with Class 2, mainly at the top (X1 between about -1 and 0, X2 between about 2 and 3).
- Class 2: The Class 2 region covers the left side of the feature space (roughly X1 $<$ -1 to 0, depending on X2) and wraps above and below the Class 3 region. The model predicts Class 2 there with probabilities close to 1. Most Class 2 observations are correctly classified, including the arcs above and below the center. The exceptions are a handful of points near the white transition band at the top (X1 between about 0 and 1, X2 between about 2 and 3) and a few that spill into the Class 3 region.
- Class 3: The Class 3 region is a compact area in the center of the feature space (roughly X1 between -2 and 2, X2 between -1.5 and 1.5). Most Class 3 observations fall within it, with the misclassified ones mostly along its edges, in the Class 2 region on the left and the Class 1 region on the right.

Probabilities within the region are high, though somewhat lower than for Class 1 and Class 2, and new observations would likely behave similarly.

Appendix

Question 1 code

```
lq=function(x,q){
  coeff=2**q
  summands_store=c()
  for(k in 0:q) { # These are the ks we are summing over

   summands_k=(x**k)*(choose(q,k)*choose(((q+k-1)/2),q)
 ↪  )
   summands_store=c(summands_store,summands_k)
  }

  f_x=coeff*sum(summands_store)

  return(f_x)

}




# plot the above
# USE sapply to vectorize the calculation across a
 ↪  supplied vector
x_list=list(x=seq(-1,1,0.1))

x=seq(-1,1,0.1)
Qs=seq(0,5,1)

q_cols=c('blue','red','magenta','lightblue','purple','green')



par(mfrow = c(2, 3),oma = c(0, 0, 3, 0))

for(i in 1:length(Qs)){
 ↪  #c(min(sapply(x,lq,q=5)),max(sapply(x,lq,q=5)))
```

```
  q_i=Qs[i]

   ↪  plot(x,sapply(x,lq,q=q_i),col=q_cols[i],lwd=1.2,xlab='x',ylab='Lq(x)',mai
   ↪  paste('Lq(x) q=',q_i),type='l')


}

mtext("Legendre Polynomials of order q", side = 3, outer
 ↪  = TRUE, cex = 1.5, font = 2)

#ya
# plots


#| label: fig-subplots
#| fig-cap: "f(x) two ways"
#| fig-subcap:
#|   - "f(x): Equation 2"
#|   - "f(x): Equation 3"
#| layout-ncol: 2


set.seed(30)

# We hace Qf coefficients


target_alpha=function(x,alpha_q,Qf){

  store_q=c()
  for(q_i in 0:Qf){

    store_q=c(store_q, (alpha_q[q_i+1]*(x**q_i)))
  }

  ret=sum(store_q)
  return(ret)
}



target_beta=function(x,beta_q,Qf){
```

```
  store_q=c()
  for(q_i in 0:Qf){
    Lq=lq(x,q_i)
    store_q=c(store_q,beta_q[q_i+1]*Lq)
  }

  ret=sum(store_q)
  return(ret)
}


# Plot the above target_alpha

plot(NULL,xlim=c(-1,1),ylim=c(-5,5),xlab='x',ylab='f(x)',main='Equation
 ↪  2 Random Target Functions')

q_cols=c('blue','red','magenta','lightblue','purple','green')

#draw aplha matrix
num_draws=3

for(i in 1:3){
  Qf=5
  alpha_q_i=runif(Qf + 1, -1, 1)

  lines(x,sapply(x,target_alpha,alpha_q=alpha_q_i
   ↪  ,Qf=5),col=q_cols[i],lwd=1.2)
}


legend("bottomleft",
       legend = c('Target 1','Target 2','Target 3'), #
        ↪  Text labels
       col = q_cols[1:3],
       pch = 1,
       cex=0.85,
       title = "Equation 2",
       bty='n')

# Plot the above target_beta


plot(NULL,xlim=c(-1,1),ylim=c(-2,2),xlab='x',ylab='f(x)',main='Equation
 ↪  3 Random Target Functions')
```

```
for(i in 1:3){
  Qf=5
  beta_q_i <- runif(Qf + 1, -1, 1)

   ↪  lines(x,sapply(x,target_beta,beta_q=beta_q_i,Qf=5),col=q_cols[i],lwd=1.2)
}


legend("topleft",
       legend = c('Target 1','Target 2','Target 3'), #
        ↪  Text labels
       col = q_cols[1:3],
       pch = 1,
       cex=0.85,
       title = "Equation 3",
       bty='n')
```

Question 3 code

```
set.seed(2023)
f_x=function(x,sig=0){
  epsi=rnorm(1,0,sig)

  if(x<0){
    target=abs(x+1)-0.5
    return (target+epsi)

  }
  if(x>=0){
    target=abs(x-1)-0.5

    return(target+epsi)

  }
}

# Simulte 5 time
f_x=function(x,sig=0){
  epsi=rnorm(1,0,sig)

  if(x<0){
    target=abs(x+1)-0.5
```

```
    return (target+epsi)

  }
  if(x>=0){
    target=abs(x-1)-0.5

    return(target+epsi)

  }
}

N=5
xg=seq(-2, 2, length.out = 400)   # grid for plotting
cols=c("darkgreen", "blue", "orange", "red", "purple")

plot(xg, sapply(xg,f_x), type = "l", lwd = 3, ylim =
 ↪  c(-2, 2),   # widen if curves are clipped
     xlab = "x", ylab = "y", main = "Target function, 5
      ↪  draws (N = 5, sigma = 0)")

set.seed(2023)
for (i in 1:5) {
  xi=runif(N, -2, 2)
  y=sapply(xi,f_x)
  points(xi, y, pch = 19, col = cols[i])

  h1=lm(y ~ xi)
  lines(xg, predict(h1, data.frame(xi = xg)), col =
   ↪  cols[i], lty = 1)

  z1=sin(pi * xi); z2=cos(pi * xi)
  h2=lm(y ~ z1 + z2 - 1)
  lines(xg, predict(h2, data.frame(z1 = sin(pi * xg), z2
   ↪  = cos(pi * xg))),
        col = cols[i], lty = 2)
}

legend("topright", legend = c("f(x)", "H1 (solid)", "H2
 ↪  (dashed)"),
       lty = c(1, 1, 2), lwd = c(3, 1, 1), bty = "n")


# Funtion Arg
```

```r
# f_tru: True DGP
# hyp: Hypothesis function
# sig: nr of std deviation from mean
# N: size of data
# nsim= nr of simulations/draws


fit_h = function(hyp, x, y){
  if(hyp == 'h1') fit = lm(y ~ x)
  if(hyp == 'h2') fit = lm(y ~ sin(pi * x) + cos(pi * x)
   ↪  - 1)
  return(fit)
}

oos_estimate = function(f_true, hyp, sig, N, nsim =
 ↪  1000, n_grid = 200){

  f_vec  = function(x) sapply(x, f_true)       # works
 ↪  even if f_true isn't vectorised
  x_grid = seq(-2, 2, length.out = n_grid)     # fixed
 ↪  evaluation points
  f_grid = f_vec(x_grid)

  pred_store    = matrix(NA, nrow = n_grid, ncol = nsim)
  oos_err_store = numeric(nsim)

  for(i in 1:nsim){
    # training data with noise
    xi = runif(N, -2, 2)
    y  = f_vec(xi) + rnorm(N, 0, sig)
    h  = fit_h(hyp, xi, y)

    # predictions on the fixed grid (column name must be
     ↪  x)
    pred_store[, i] = predict(h, data.frame(x = x_grid))

    # fresh OOS sample with noise
    xo = runif(N, -2, 2)
    yo = f_vec(xo) + rnorm(N, 0, sig)
    oos_err_store[i] = mean((predict(h, data.frame(x =
 ↪  xo)) - yo)^2)
  }
```

```
  g_bar      = rowMeans(pred_store)                    #
 ↪  mean prediction at each grid x
  model_bias2 = mean((g_bar - f_grid)^2)
  model_var   = mean((pred_store - g_bar)^2)           #
 ↪  g_bar recycles down the columns
  both        = model_bias2 + model_var + sig^2        #
 ↪  noise variance = sig^2
  oos_err_est = mean(oos_err_store)
  estimate_diff = abs(both - oos_err_est)

  return(list(model_bias2 = model_bias2, model_var =
   ↪  model_var, both = both,
              oos_err_est = oos_err_est, estimate_diff =
               ↪  estimate_diff))
}

# Test
library(dplyr); library(tibble); library(knitr)
set.seed(321)
f_true = f_x
sig = 0
N = 5

h1_out = oos_estimate(f_true, hyp = 'h1', sig, N, nsim =
 ↪  1000)
h2_out = oos_estimate(f_true, hyp = 'h2', sig, N, nsim =
 ↪  1000)

comparison_table = rbind(data.frame(h1_out),
 ↪  data.frame(h2_out)) %>%
  mutate(model = c('h1(x)', 'h2(x)')) %>%
  column_to_rownames(var = "model")

kable(comparison_table, digits = 4,
      col.names = c('Bias', 'Variance',
       ↪  'Both:Bias+Var+Noise', 'OOS Error', 'Estimate
       ↪  Difference'),
      align = "cc")
```

Question 4 code

```
dat = read.table('Collider_Data_2022.txt', h =
 ↪  TRUE,stringsAsFactors =T)
```

```
plot(NULL,
 ↪  xlim=c(min(dat['X1']),max(dat['X1'])),ylim=c(min(dat['X2']),max(dat['X2'])),
 ↪  xlab='X1',ylab='X2',main='Classification Scatter
 ↪  Plot')

points(dat[dat$Yi1==1,]$X1,dat[dat$Yi1==1,]$X2,col='blue',pch=16)

points(dat[dat$Yi2==1,]$X1,dat[dat$Yi2==1,]$X2,col='red',pch=16)

points(dat[dat$Yi3==1,]$X1,dat[dat$Yi3==1,]$X2,col='purple',pch=16)

legend(x='topleft',
       legend = c('Class 1', 'Class 2', 'Class 3'),
       col=c('blue','red','purple'),
       pch=c(16,16,16),
       title = "Groups",
       cex=0.85,
       bty = "o")


soft_max = function(z) {  # assume z is a matrix

  z_shifted = sweep(z, 2, apply(z, 2, max), FUN = "-") #
 ↪  subtract maximum from each column to avoid overflow

  num = exp(z_shifted)

  # Sum exponentials within each column
  sums = colSums(num)

  # Normalize each column
  result = sweep(num, 2, sums, FUN = "/")

  return(result)
}



X_feat=cbind(dat$X1,dat$X2)
Y=cbind(dat$Yi1,dat$Yi2,dat$Yi3)
```

```
feedforward_nn = function(X_input, y_input, theta, m,
 ↪  nu,
                           activation_fun_2 = soft_max,
                            ↪  activation_fun_1 = tanh){

 y  = as.matrix(y_input)
 N  = nrow(X_input); p = ncol(X_input); d2 = ncol(y)
 X  = t(as.matrix(X_input))                          # p x
↪  N

 # split theta: W1, W2, b1, b2
 i1 = p * m
 i2 = i1 + m * d2
 i3 = i2 + m
 W1 = matrix(theta[1:i1],          nrow = p, ncol = m)
 W2 = matrix(theta[(i1 + 1):i2],   nrow = m, ncol = d2)
 b1 = theta[(i2 + 1):i3]
 b2 = theta[(i3 + 1):(i3 + d2)]

 # forward pass
 first_layer_comp = activation_fun_1(sweep(t(W1) %*% X,
↪  1, b1, "+"))       # m x N
 second_layer_comp_no_activation = sweep(t(W2) %*%
↪  first_layer_comp, 1, b2, "+")  # d2 x N
 second_layer_comp =
↪  activation_fun_2(second_layer_comp_no_activation)
↪  # d2 x N

 # one-hot predicted labels (d2 x N): max class within
  ↪  each observation
 y_pred_label = apply(second_layer_comp, 2,
                      function(pr)
                       ↪  as.numeric(seq_along(pr) ==
                       ↪  which.max(pr)))

 # cross-entropy: sum over classes and observations,
  ↪  divided by N
 CE_error    = -sum(t(y) * log(pmax(second_layer_comp,
↪  1e-12))) / N
 CE_error_L2 = CE_error + (nu / N) * (sum(W1^2) +
↪  sum(W2^2))

 return(list(first_layer_comp = first_layer_comp,
```

```
              second_layer_comp = second_layer_comp,
              second_layer_comp_no_activation =
               ↪  second_layer_comp_no_activation,
              y_pred_label = y_pred_label,
              CE_error = CE_error,
              CE_error_L2 = CE_error_L2))
}
set.seed(2022)

y_input=Y
X_input=X_feat
#
m   = 10 # number nodes in hidden layer
p     = dim(X_input)[2]
q     = dim(as.matrix(Y))[2]

npars = p*m+m*q+m+q
theta_rand = runif(npars,-1,1)
N       = dim(X_feat)[1]
set     = sample(1:N,0.5*N,replace = FALSE) #50 50 split
X_train = X_feat[set,,drop = FALSE] #
Y_train = Y[set,,drop = FALSE]
X_val   = X_feat[-set,,drop = FALSE]
Y_val   = Y[-set,,drop = FALSE]

obj_pen = function(pars)
{
  res_model =
 ↪  feedforward_nn(X_train,Y_train,pars,m,nu,activation_fun_2=soft_max,activati
 ↪
  return(res_model$CE_error_L2) # Reg Obj
}

n_nus = 15
nu_seq= exp(seq(-2,2,length = n_nus)) # sequence of nus
 ↪  to validate against
val_error = rep(0,n_nus)
for(i in 1:n_nus)
{
    nu = nu_seq[i] # Pick nu
    res_opt = nlm(obj_pen,theta_rand, iterlim = 1000) #
 ↪  Fit for that nu
```

```
    #Evaluate on Validation Set with optimal parameters
     ↪  and particular nu

    res_val=
 ↪  feedforward_nn(X_val,Y_val,res_opt$estimate,m,nu,activation_fun_2=soft_max,

    # Get reg validation Error
  val_error[i] = res_val$CE_error

  #print(paste0('Validation run:',i,' nu = ', nu))
}


#| fig-dpi: 300


min_val_error=min(val_error)



best_nu_index=which(val_error==min(val_error))
best_nu=nu_seq[best_nu_index];


# Compare the error of the nu the right of the best

next_val_error=val_error[(best_nu_index+1):length(val_error)]

next_seq=nu_seq[(best_nu_index)+1:length(nu_seq)]

next_best_nu=next_seq[which.min(next_val_error)] ;

next_min_val_error=min(next_val_error)




plot(val_error~log(nu_seq), type = 'b',pch =
 ↪  16,xlim=c(min(log(nu_seq))-1,max(log(nu_seq))+1),
 ↪  main='Validation CE Error vs Regularization
 ↪  Parameter (nu)', xlab='lambda
 ↪  (log)',ylab='validation error')
```

```
abline(v =log(best_nu), col = "purple", lty = "dashed",
 ↪  lwd = 2)

abline(v =log(next_best_nu), col = "blue", lty =
 ↪  "dashed", lwd = 2)

text(x = log(best_nu)-0.5, y = 0.5, labels = paste("best
 ↪  nu:",round(best_nu,3), 'Val CE
 ↪  Error:',round(min_val_error,2),sep = "\n"),
     srt = 0, adj = c(0.5, -0.5), col =
      ↪  "purple",cex=0.68)


text(x = log(next_best_nu)+0.5, y = 0.5, labels =
 ↪  paste("next best nu:",round(next_best_nu,3), 'Val
 ↪  Error:',round(next_min_val_error,2),sep = "\n"),
     srt = 0, adj = c(0.5, -0.5), col = "blue",cex=0.68)
#
#


library(colorspace)

# Fit probability color mapping for each class onto the
 ↪  feature space



# Helper functions

# Assign color pallet based on indexing
  ass_pallet=function(ind_val,col_steps=50){
    if(ind_val==1){

      # Class 1, high to low prob color mapping
      pallet=colorRampPalette(c("lightblue",
 ↪  'white',"darkblue"))

    }

    if(ind_val==2){

      # Class 2
```

```
      pallet=colorRampPalette(c("mistyrose",'white',
 ↪  "darkred"))

    }
   if(ind_val==3){

     # Class 3
      pallet=colorRampPalette(c("mediumpurple1",'white',
 ↪  "purple4"))

   }

    return(pallet(col_steps))
  }


# Assign a color to prediction based on its class and
 ↪  strength of probability value

col_pred=function(x,col_steps=50){
  # x=prediction matrix from ff nn

  max_ind=apply(x,2,which.max) #get index of max
 ↪  probability
  max_prob=apply(x,2,max) # get value of max probability
  col_range=seq(0,1,length.out = col_steps)
  col_pallet=sapply(max_ind,ass_pallet) #prob colors for
 ↪  each class, matrix
  color_indices= findInterval(max_prob, col_range) # for
 ↪  each obs, find its indx in a range of 0 to 1

  col_pallet_N=ncol(col_pallet)
  selected_cols= col_pallet[cbind(color_indices,
 ↪  seq_len(col_pallet_N))] # get appropriate index in
 ↪  the pallette assignment matrix

  return(selected_cols)
}

pal1 = ass_pallet(1)
pal2 = ass_pallet(2)
pal3 = ass_pallet(3)
```

```
obj_pen = function(pars)
{
  res_model
 ↪  =feedforward_nn(X_input,y_input,pars,m,next_best_nu,activation_fun_2=soft_m
  return(res_model$CE_error_L2) # Reg Obj
}

# Optimize weights using next best nu
res_opt_best_nu = nlm(obj_pen,theta_rand, iterlim =
 ↪  1000)



M   = 160
x1  = seq(-5,+5,length = M)
x2  = seq(-5,+5,length = M)
xx1 = rep(x1,M)
xx2 = rep(x2,each = M)

layout(matrix(1:4, nrow = 1),
       widths = c(5, 0.7, 0.7, 0.7))

par(mar = c(4, 4, 3, 1))

plot(NULL,
 ↪  xlim=c(-5,5),ylim=c(-5,5),xlab='x1',ylab='x2',
 ↪  main='Feed Forward NN Response Curve')

abline(h= x2,v = x1, lwd = 0.3,col='black')




XX = cbind(xx1,xx2)
YY1 = rep(1,M^2)
YY2=rep(1,M^2)
YY3=rep(1,M^2)
YY=cbind(YY1,YY2,YY3)


# Again use next best nu
res_fitted=feedforward_nn(XX,YY,res_opt_best_nu$estimate,m,next_best_nu,activat
 ↪
```

```
points(xx2~xx1,pch = 16,
 ↪  col=col_pred(res_fitted$second_layer_comp) ,cex = 2)

points(dat[dat$Yi1==1,]$X1,dat[dat$Yi1==1,]$X2,col='blue',pch=10)

points(dat[dat$Yi2==1,]$X1,dat[dat$Yi2==1,]$X2,col='red',pch=10)

points(dat[dat$Yi3==1,]$X1,dat[dat$Yi3==1,]$X2,col='purple',pch=10)

add_colour_bar = function(pal, class_name) {

  par(mar = c(4, 0.5, 3, 2))

  probs <- seq(0, 1, length.out = length(pal))

  # Draw continuous-looking gradient
  z = matrix(rep(probs, each = 2), nrow = 2)

  image(x = c(0, 1),
        y = probs,
        z = z,
        col = pal,
        zlim = c(0, 1),
        axes = FALSE,
        xlab = "",
        ylab = "",
        main = class_name)

  axis(4, at = seq(0, 1, by = 0.2),
       las = 1, cex.axis = 0.75)

  box()
}

#colour mapping
add_colour_bar(pal1, "Class 1")
add_colour_bar(pal2, "Class 2")
add_colour_bar(pal3, "Class 3")
```